Forecasting is a scientific process which aims at reducing the uncertainty of the future state of business and trade, not dependent merely on guess work, but with a sound scientific footing for the decision on the future course of action.

NOTE

1. Long term forecasting can be found using Trend.
2. Short term forecasting can be found using seasonal variations.

7.4.1 Definition

"Forecasting refers to the analysis of past and present conditions with a view of arriving at rough estimates about the future conditions.

According to T.S. Lewis and R.A. Fox "Forecasting is using the knowledge we have at one time to estimate what will happen at some future moment of time".

Forecasting is an important tool that serves many fields including business and industry, government, economics, environmental sciences, medicine, social science, politics and finance. Forecasting problems are often classified as short-term, medium-term, and long-term.

Short-term forecasting problems involve predicting events for a few time periods (days, weeks, months) into the future.

Medium-term forecast extends from one to two years into the future.

Long-term forecasting problems can extend beyond that by many years.

Short and medium-term forecasts are required for activities that range from operations management to budgeting and selecting new research and development projects. Long term forecasts impact issues relating to strategic planning.

POINTS TO REMEMBER

- Time series is a time oriented sequence of observations.
- Components of time series are secular trend, seasonal variations, cyclical variations and irregular (erratic) variations
- Methods of calculating trend values are graphical method, semi - averages method, moving averages method, and method of least squares.
- The line $y = a + b\,x$ found out using the method of least squares is called 'line of best fit'.
- Normal equations involved in the method of least squares are

$$\sum_{i=1}^{n} y_i = na + b\sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i y_i = a\sum_{i=1}^{n} x + b\sum_{i=1}^{n} x_i^2$$

- Seasonal indices may be found out by using simple average method.
- Forecasting is the analysis of using past and present conditions to get rough estimates of the future conditions
- Forecasting methods can be short-term, medium-term and long-term.

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